

LECTURE 6

Reinforcement Learning Part 1: Q-Learning & Deep Q-Networks

MDPs • Bellman equations • Q-learning • ε-greedy • DQN • replay buffer • reward shaping

Case study: wargame tactical agent, 20×20 JANUS grid, 100K episodes, 81% win vs novice human

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AI & Machine Learning

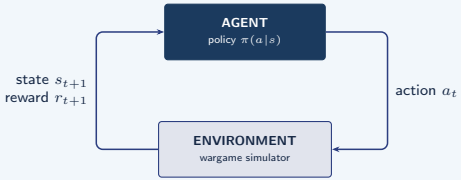
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The Third Paradigm — Learning by Doing: No Labels, Just Rewards

SUPERVISED
Labels (x, y); loss $L(f(x), y)$; static labelled dataset; immediate feedback per sample.
Example: radar classification.
Weakness: needs labels; can't exceed the training distribution.

UNSUPERVISED
No labels $\{x\}$; reconstruction / likelihood loss; feedback: none, data-driven.
Example: OPFOR clustering.
Weakness: no goal direction; can't optimise a specific objective.

REINFORCEMENT
Trial and error; loss = negative cumulative reward; data generated by interaction; delayed, sparse feedback.
Example: wargame tactical agent.
Strength: learns strategy from experience.



Why RL for defence? In wargaming and autonomy we cannot label the "correct" decision for every battle state (there are $\sim 10^{28}$). RL trains by playing millions of simulated games, learning strategic patterns no analyst could pre-specify — and no human has to label good vs bad moves.

Markov Decision Process — Five Elements Define the Agent's World

(S, A, P, R, γ)

S — **state space:** unit positions, resources, OPFOR positions, terrain, weather, ammo, morale: 20×20 grid $\times$ resources $\times$ OPFOR $\approx 10^{28}$ states.

A — **action space:** {move N, S, E, W, attack, defend, resupply} = 7 per unit; 8 units $\Rightarrow 7^8 = 5.8$M joint options per turn.

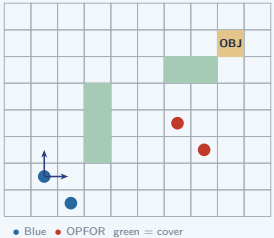
$P(s'|s, a)$ — **transitions:** set by the combat-resolution model (terrain, weather, unit quality).

$R(s, a, s')$ — **reward:** $+10$ capture objective, $+5$ destroy enemy unit, -8 lose friendly unit.

γ — **discount:** weight on future reward.

RETURN TO MAXIMISE

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$



Discount intuition: with $\gamma = 0.95$ a reward 14 turns away is worth $0.95^{14} \approx 0.49$ of an immediate one — the agent cares about the future, but not infinitely.

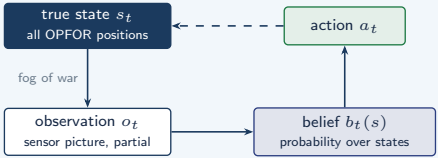
The Markov Property — and Why Real Warfare Is a POMDP

MARKOV PROPERTY

$$P(s_{t+1} \mid s_t, a_t, s_{t-1}, a_{t-1}, \ldots) = P(s_{t+1} \mid s_t, a_t)$$

- The future depends only on the **present** state, not the whole history
- If it holds, the agent need not remember all past decisions — the current state suffices for optimal action
- So the state must encode EVERYTHING relevant: ammo, morale, previous attrition, known OPFOR dispositions, intelligence quality

If relevant information is missing from the state, the problem becomes a **POMDP** (partially observable MDP) — harder: the agent must maintain a *belief* over the true state. Real warfare is almost certainly a POMDP.



Simplification: most wargame RL systems assume full observability (known OPFOR positions) — a major assumption that limits real-world transfer. Lecture 8 returns to partial observability for multi-agent teams.

Bellman Equations — Recursive Characterisation of the Optimal Policy

BELLMAN OPTIMALITY

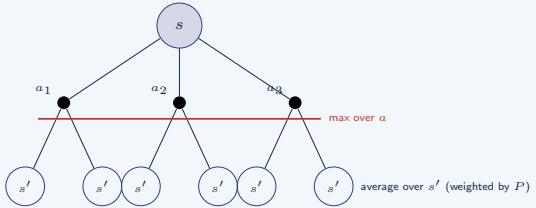
$$V^*(s) = \max_a \sum_{s'} P(s'|s,a)\big[R(s,a,s') + \gamma V^*(s')\big]$$

$$Q^*(s,a) = \sum_{s'} P(s'|s,a)\big[R(s,a,s') + \gamma \max_{a'} Q^*(s',a')\big]$$

$$V^*(s) = \max_a Q^*(s,a), \qquad \pi^*(s) = \arg\max_a Q^*(s,a)$$

- "The value of s = the best action's expected immediate reward PLUS discounted future value"
- Policy evaluation: $V^\pi(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a)[R + \gamma V^\pi(s')]$; V^* = max over policies
- If we know Q^*, the optimal policy is trivial: take $\arg\max_a Q^*(s,a)$ — the goal of Q-learning

Backup diagram

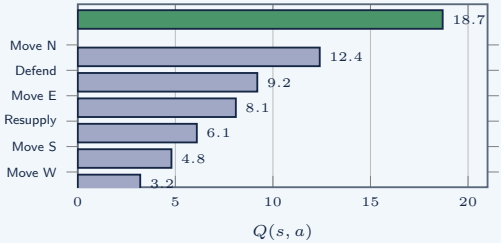


Value flows back up the tree: average over what the environment does, maximise over what the agent does.

Wargame Worked Example — Reading Q-Values

State s

Blue unit adjacent to enemy; ammo HIGH; morale HIGH; terrain COVER; objective nearby. Q-values from the trained model:



Optimal action: $\pi^*(s) = \arg\max_a Q =$ **ATTACK** (18.7).

What the numbers mean

$Q^*(s, \text{Attack}) = 18.7$: expected cumulative *discounted reward* from s if I attack now and then play optimally from every later state. $Q^*(s, \text{Defend}) = 9.2$: defending yields less total reward in this specific state (high ammo, in cover).

Interpretation pitfall: Q-values are in *reward units*, not probabilities. "18.7" does not mean "18.7% chance of winning".

Q-Learning (Watkins, 1989) — Model-Free, Off-Policy TD Control

TD UPDATE RULE

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[\underbrace{r + \gamma \max_{a'} Q(s',a')}_{\text{TD target}} - Q(s,a)\Big]$$

$$\delta = r + \gamma \max_{a'} Q(s',a') - Q(s,a) \quad \text{(TD error)}$$

- **Model-free:** no need to know $P(s'|s,a)$ or R in advance; learns Q^* from interaction
- $\alpha \in (0,1]$ learning rate; "shift the estimate toward the target by a fraction α of the error"
- **Off-policy:** uses $\max_{a'}$ (the greedy next action) regardless of the action actually taken next

Numerical walk-through ($\alpha = 0.1$, $\gamma = 0.9$)



One update

TD target $= 0 + 0.9 \times 14.2 = 12.78$; TD error $\delta = 12.78 - 5.0 = 7.78$ (positive: MOVE N was undervalued because it leads to a high-value state). New $Q(s_1, \text{N}) = 5.0 + 0.1 \times 7.78 = \mathbf{5.778}$. After 1,000 episodes ≈ 11.8, converging toward 12.78.

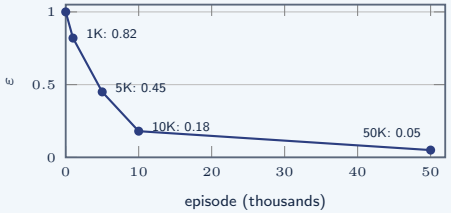
ε-Greedy Exploration — the Exploration–Exploitation Dilemma

ε-GREEDY POLICY AND ANNEALING

$$P(a = a^*) = (1 - \varepsilon) + \frac{\varepsilon}{|A|}, \qquad P(a \neq a^*) = \frac{\varepsilon}{|A|}$$

$$\varepsilon_t = \varepsilon_{\min} + (\varepsilon_{\max} - \varepsilon_{\min})\, e^{-t/\tau}$$

- With probability ε: EXPLORE (random action); otherwise EXPLOIT ($\arg\max Q$)
- Start $\varepsilon = 1.0$ (know nothing); end $\varepsilon = 0.05$ (mostly exploit, keep 5% exploring); typical wargame $\tau = 5{,}000$ episodes



Case-study schedule (as reported).

A commander faces the same dilemma: keep using the tactic that has worked, or probe an untried approach that might work better. Never exploring locks in a sub-optimal habit.

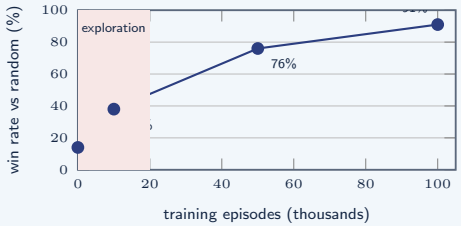
Q-Learning Convergence — the Theory and the Tabular Wall

Theorem (Watkins & Dayan, 1992)

Q-learning converges to Q^* with probability 1 if: ① every (s, a) is visited infinitely often ($\varepsilon > 0$); ② learning rates satisfy Robbins–Monro: $\sum_t \alpha_t = \infty$, $\sum_t \alpha_t^2 < \infty$ (e.g. $\alpha_t = 1/t$); ③ stationary rewards; ④ finite state–action space.

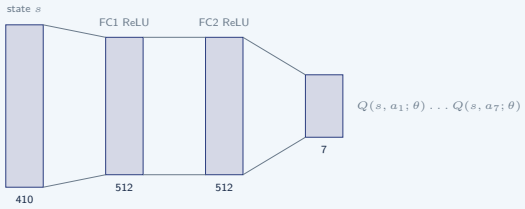
In practice: a small constant α (e.g. 0.001) — theoretically non-convergent, but it works well and adapts to non-stationary environments.

Tabular limitation: Q-table size $= |S| \times |A|$. For the wargame $10^{28} \times 7$ — impossible to store or enumerate. Solution: **function approximation** — replace the table with a neural network $Q(s, a; \theta)$: the DQN.



DQN on the 20×20 grid (as reported: 10K 38%, 50K 76%, 100K 91%); the episode-0 point is illustrative. Phases: 0–20K random-like, 20–60K rapid improvement, then refinement.

Deep Q-Network — Replace the Q-Table with a Neural Network



- Input: 400-dim grid encoding + unit attributes (ammo, HP, morale) = 410 features
- One forward pass → all 7 Q-values; parameters $410 \cdot 512 + 512 \cdot 512 + 512 \cdot 7 = 479{,}744$
- Adam, lr 0.0001 (Mnih et al., 2015 style)

DQN LOSS

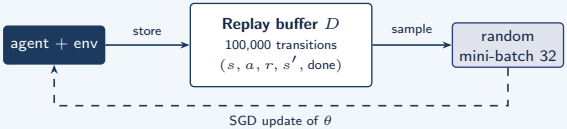
$$L(\theta) = \mathbb{E}\big[(y - Q(s,a;\theta))^2\big]$$

$$y = r + \gamma \max_{a'} Q(s',a';\theta^-)$$

- θ: online network, trained by SGD
- θ^-: **target network**, a frozen copy used only to compute y

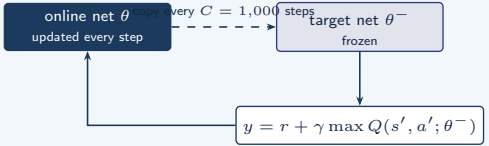
Generalisation is the whole point: states never visited still get sensible Q-values because similar states share network weights — impossible with a table.

DQN's Two Key Innovations — Experience Replay and the Target Network



Innovation 1: experience replay

① breaks temporal correlation between consecutive samples (correlated updates cause divergence); ② each experience is reused ∼1,000 times (sample efficiency); ③ stabilises training dramatically.



Innovation 2: the target network

Without it, the target y moves every time θ changes — chasing a moving target ⇒ instability. Frozen θ^- keeps the target stable for C steps.

Without both innovations, DQN often diverges (vanilla DQN: 74% vs random and unstable).

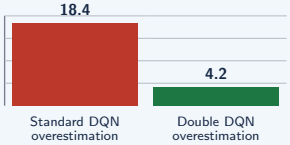
Double DQN and Dueling DQN — Better Value Estimates

Double DQN (van Hasselt 2016): fix overestimation

DECOUPLE SELECTION FROM EVALUATION

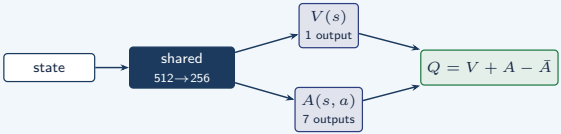
$$a^* = \arg\max_{a'} Q(s', a'; \theta), \qquad y = r + \gamma Q(s', a^*; \theta^-)$$

Standard DQN's $\max$ picks the noisiest estimate, so Q-values are systematically too high and the agent becomes over-confident on rare states. One network picks; the other scores.



Average overestimation after 50K episodes (reward units).

Dueling DQN (Wang 2016): value + advantage



DECOMPOSITION

$$Q(s,a) = V(s) + A(s,a) - \frac{1}{|A|}\sum_{a'} A(s,a')$$

V: how good is the state regardless of action? A: how much better is a than average? Subtracting the mean makes V and A identifiable. Helps in states where all moves are similar (no enemy near).

Reward Shaping — When +100 Win / −100 Lose Is Too Sparse to Learn From

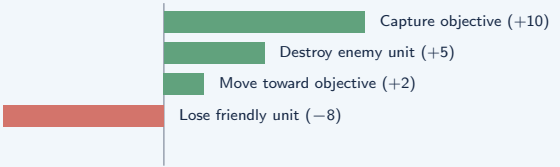
Naive reward: $+100$ win, -100 lose, 0 otherwise. Games last $\sim$40 turns, so 40 actions get no feedback; early on the agent almost always loses — all -100s, no gradient. Result: a random policy for the first 20,000+ episodes.

POTENTIAL-BASED SHAPING (NG, HARADA & RUSSELL 1999)

$$R_{\text{shaped}} = R(s,a,s') + F(s,s'), \qquad F = \gamma\Phi(s') - \Phi(s)$$

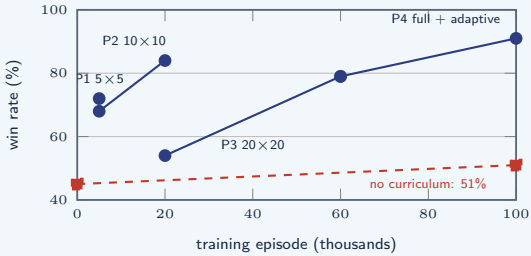
$\Phi(s)$ estimates "how good is s". Potential-based shaping adds guidance WITHOUT changing the optimal policy.

Wargame shaping rewards



Danger: badly designed shaping teaches the agent to farm the bonus (e.g. circling near the objective to collect "move toward" rewards) instead of winning. Keep shaping potential-based and audit agent behaviour.

Curriculum Learning — Start Simple, Add Complexity



Each phase transfers weights to the next; the drops at phase starts are the harder scenario, not forgetting.

The no-curriculum start value is illustrative.

The wargame curriculum

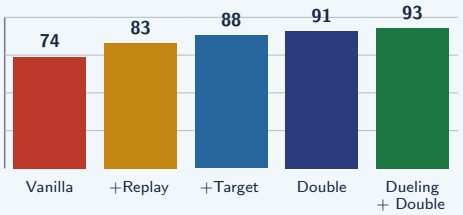
P1 (0–5K): 5×5, 1 v 1, fixed positions; a Q-table works: 72%. **P2 (5–20K):** 10×10, 3 v 3, two terrain types; transfer to DQN: 68%→84%. **P3 (20–60K):** 20×20, 8 v 8, 5 terrains + weather: 54%→79%. **P4 (60–100K):** full + adaptive OPFOR: 79%→91% vs random; 63%→81% vs novice human.

Key result: trained directly on the full scenario, DQN reached only **51%** after 100K episodes vs **91%** with the curriculum — a 40-point gain. Like cadets: drills before battles.

Wargame Case Study — DQN Agent on the 20×20 JANUS Hex Grid

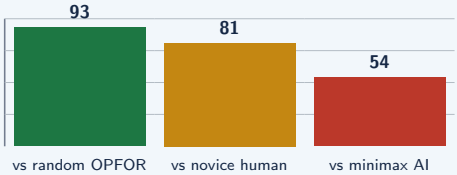
Algorithm	Opponent	Win	Turns	Notes
Tabular Q (10×10)	random	91%	22.4	table infeasible at 20×20
DQN (vanilla)	random	74%	31.8	no replay/target: unstable
DQN + replay	random	83%	28.2	replay alone stabilises
DQN + replay + target	random	88%	26.7	stable convergence
Double DQN	random	91%	25.1	overestimation corrected
Dueling + Double	random	**93%**	23.8	best vs random
Dueling + Double	novice human	81%	29.4	human adapts
Dueling + Double	minimax AI	54%	38.2	classical AI competitive

All DQN variants: 100K training episodes; tabular Q: 50K.



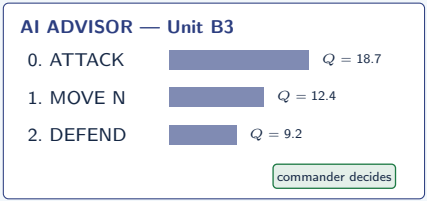
Win rate vs random OPFOR (%).

Each stabilisation trick adds measurable performance — but the opponent matters more than the algorithm: 93% → 81% → 54%.



Sim-to-real warning: 91% vs random OPFOR in simulation $\neq$ 91% in a field exercise. Human OPFOR adapts, introduces friction, and exploits the AI's pattern repetition. Minimax at 54% shows classical adversarial search remains highly competitive — DQN is not a solved problem for wargame AI.

Recommended role: ADVISOR, not commander



Mock-up: top-3 actions with Q-values; the human selects.

Operational insight: propose the top-3 actions with their Q-values and let the commander choose. Human + AI outperforms either alone.

Reward-driven learning was discovered in animals, formalised for the military, and proven in games

PSYCHOLOGY
Thorndike's law of effect (1898) and Skinner's operant conditioning: actions followed by reward are repeated — the behavioural root of RL.
Defence: training design — immediate, consistent feedback speeds skill acquisition, as reward shaping does for agents.

BIOLOGY / NEUROSCIENCE
Dopamine neurons fire in proportion to the *reward-prediction error* — almost exactly the TD error δ (Schultz, Dayan & Montague, *Science*, 1997).
Lesson: Q-learning's update mirrors how brains learn.

ENGINEERING / OR
Richard Bellman developed dynamic programming at the RAND Corporation in the 1950s — the Bellman equation came from defence-funded operations research.
Defence: optimal control, logistics scheduling, missile guidance.

AERONAUTICS
RL controllers learned to fly a model helicopter through aerobatic manoeuvres, including sustained inverted flight (Ng et al., Stanford, 2004).
Defence: autonomous flight control under turbulence — the theme of Lecture 7.

STATISTICS
Thompson (1933) proposed allocating treatments in clinical trials by posterior probability — the multi-armed bandit, origin of the exploration–exploitation trade-off.
Defence: sensor tasking: keep watching known emitters or search for new ones?

GAMES & WARGAMING
TD-Gammon (Tesauro, 1990s) reached expert backgammon by self-play; in DARPA's AlphaDogfight Trials (2020) an RL agent beat an experienced F-16 pilot 5–0 in simulation.
Caution: simulation victories are not combat victories.

Common thread: learn from consequences, balance exploring against exploiting, and beware of rewarding the wrong thing — in animals, organisations and agents alike.

Lecture 6 Summary — From MDP Formalism to Wargame Tactical Agents

MDP FORMALISM
(S, A, P, R, γ); Markov property: the future depends on the present; wargame $\sim 10^{28}$ states $\Rightarrow$ DQN.
First: define the problem

BELLMAN EQUATIONS
$Q^*(s,a) = \sum P[R + \gamma \max Q^*]$; $\pi^* = \arg\max Q^*$; Q-values are reward units, not probabilities.
Core: recursive optimality

Q-LEARNING
$Q \leftarrow Q + \alpha[r + \gamma \max Q' - Q]$; model-free, off-policy, convergent (tabular); sparse reward slows it.
Core: the update rule

DQN INNOVATIONS
Experience replay breaks correlations; target network stabilises targets; without both, training often diverges.
Essential: for stability

DOUBLE & DUELING
Double: overestimation 18.4 $\rightarrow$ 4.2; dueling: $V + A$ decomposition; 93% vs random, 81% vs novice.
Best: wargame performance

REWARD & CURRICULUM
Sparse ± 100 is too slow; potential-based shaping; curriculum 91% vs 51%.
Critical: for practical deployment

From choosing among 7 actions to steering continuously

- REINFORCE: optimise the policy directly
- Actor–critic methods
- PPO (Proximal Policy Optimisation)
- Continuous action spaces (thrust, heading, throttle)

Lecture 7 case study

Multi-drone coordination with continuous waypoint control and 6-DoF flight dynamics.

Assignment 6 is released with this pack. Lecture 8 then covers multi-agent RL (MARL) and autonomous combat platforms.

